Question 1

- What are the two categories of data compression methods?

Question 2

- What is the difference between lossless compression and lossy compression?

Question 3

- What is run-length encoding?

Question 4

- How does Lempel Ziv encoding reduce the amount of bits transmitted?

Question 5

- What is Huffman coding?

Question 6

- What is the role of the dictionary in LZ encoding?

Question 7

- What is the advantage of LZ encoding over Huffman coding?

Question 8

- Name three lossy compression methods.

Question 9

- Encode the following bit pattern using run-length encoding with 5-bit codes:

  18 zeros, 11, 56 zeros, 1, 15 zeros, 11

Question 10

- Encode the following characters using Huffman coding with the given frequencies:

  A (12), B (8), C (9), D (20), E (31), F (14), G (8)

Question 11

- Encode the message BAABBBBAACAA using the following Huffman code:

  A: 0 B: 10 C: 11

Question 12

- Decode the message 0101000011110 using the following Huffman code:

  A: 0 B: 10 C: 11

Question 13

- Encode the message BAABBBBAACAA using the Lempel Ziv method, then decode the encoded message to get the original message.

Question 14

- Encode the string AAAABBCCCBBB (part of a message) using the Lempel Ziv method if the dictionary contains ABB. Show the final contents of the dictionary.

Question 15

- Prove the statement stated as under:

"If a receiver knows the message being transmitted, the amount of information carried will be zero."

Question 16

- Verify the following expression:

$I(x_i x_j) = I(x_i) + I(x_j)$ if x_i and x_j are independent

Question 17

- Calculate the amount of information if it is given that $P(x_i) = ¼$.

Question 18

- A Discrete Memoryless Source (DMS) X has four symbols x_1 x_2, x_3, x_4 with probabilities

  $P(x_1) = 0.4$, $P(x_2) = 0.3$, $P(x_3) = 0.2$, $P(x_4) = 0.1$.

  (i) Calculate H(X).

  (ii) Find the amount of information contained in the messages x_1 x_2 x_1 x_3 and x_4 x_3 x_3 x_2, and compare with the H(X) obtained in part (i).

Question 19

- A DMS X has five symbols x1 , x2, x3, x4, and x5 with P(x1) = 0.4, P(x2) = 0.19, P(x3) = 0.16, P(x4) = 0.15, and P(x5) = 0.1.
- Construct a Huffman code for X, and calculate the efficiency of the code.